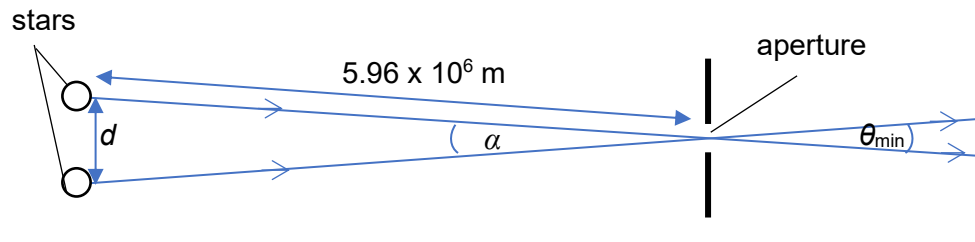


through.

17 B Using small angle approximation,



$$\alpha \approx \theta_{\min}$$

$$\frac{d_{\text{between the stars}}}{D_{\text{from telescope}}} = \frac{\lambda}{b}$$

$$\begin{aligned} d_{\text{between the stars}} &= \frac{\lambda D_{\text{from telescope}}}{b} \\ &= \frac{(630 \times 10^{-9})(5.9 \times 10^{16})}{45 \times 10^{-2}} \\ &= 8.3 \times 10^{10} \text{ m} \end{aligned}$$

18 B The distance from the node to the end of the tube is $\frac{\lambda}{2}$. Since $\frac{\lambda}{2} = 0.17 \rightarrow \lambda = 0.68 \text{ m}$. The